
Previous Title: Advanced Methods, Explainable AI, Uncertainties

Generative AI & LLMs for Environmental Science

From Language Models to AI Agents

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What Are LLMs?

The Core Idea

- Neural networks trained to predict the next token across massive text corpora
- Built on the **Transformer** architecture — self-attention
- Emergent skills at scale: reasoning, code, multimodal understanding

Why It Matters for Science

- Parse papers, write code, summarize data — in plain language
- Multimodal models read images, charts, satellite tiles
- Programmable research tools, not just chatbots



LLM as ~~Word~~ ^{Token} Continuation

autoregressive generation

- **A function with many parameters** — hundreds of millions to trillions
- **Parameters are learned from massive datasets** (training)
- **Input is a prompt; output is the next token, then next** (inference)

⚡ Emergent Intelligence

The key insight is that even complex reasoning, creative writing, or problem-solving can be viewed as very sophisticated pattern matching and text prediction - though the patterns LLM has learned are incredibly nuanced and allow for contextual, coherent, and useful responses rather than just simple word associations.

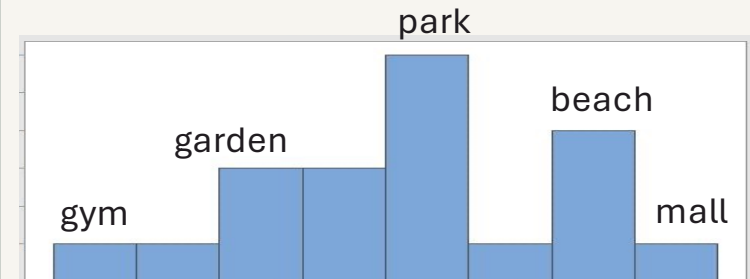
Input Text (Prompt):

"The weather today is sunny, so I think I'll go to the"

LLM Analyzes Patterns:

- "go to the" → usually followed by locations
- "sunny weather" → suggests outdoor activities
- Looking for most probable continuations

Predicted Continuations:



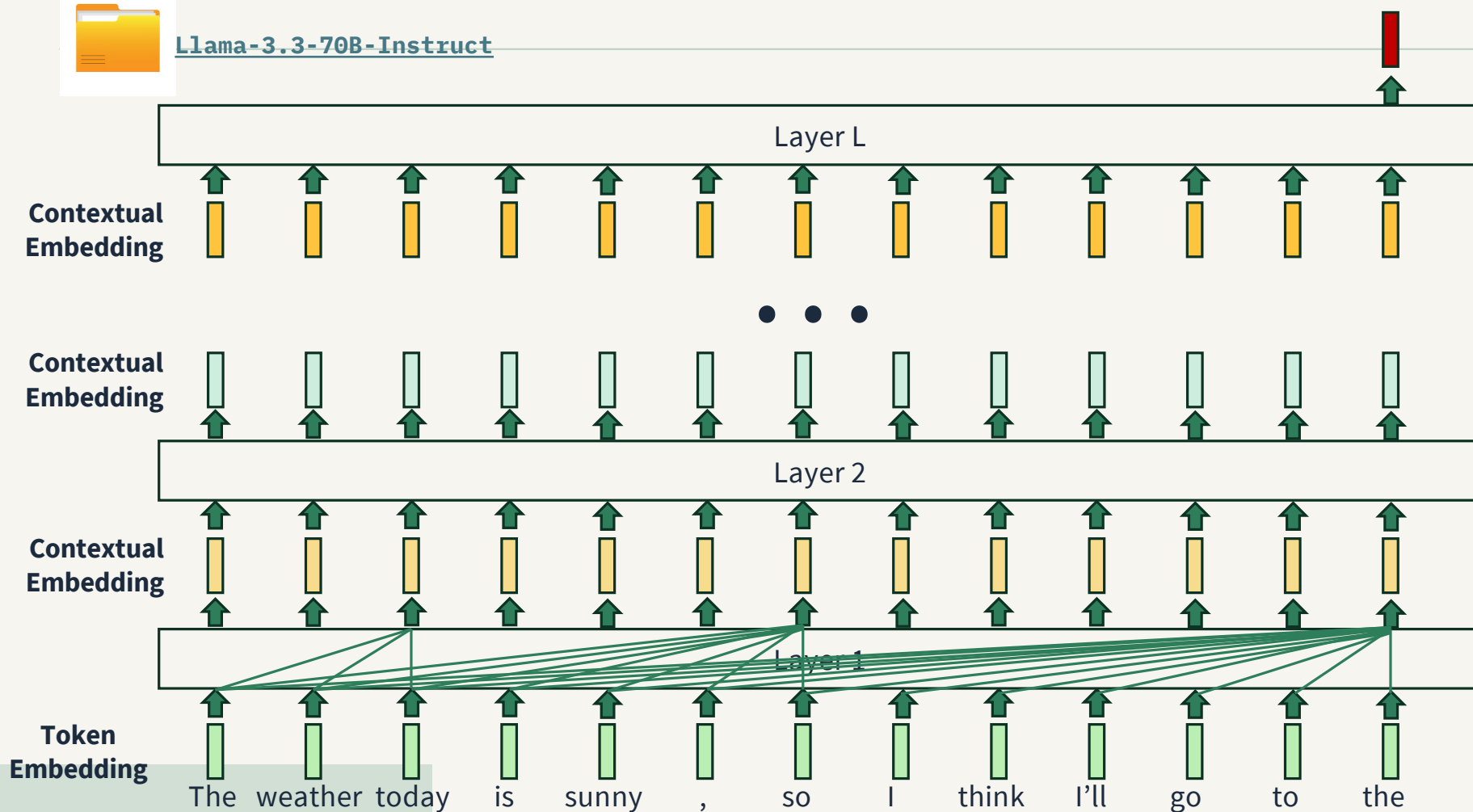
Selected Output:

"The weather today is sunny, so I think I'll go to the **park**"

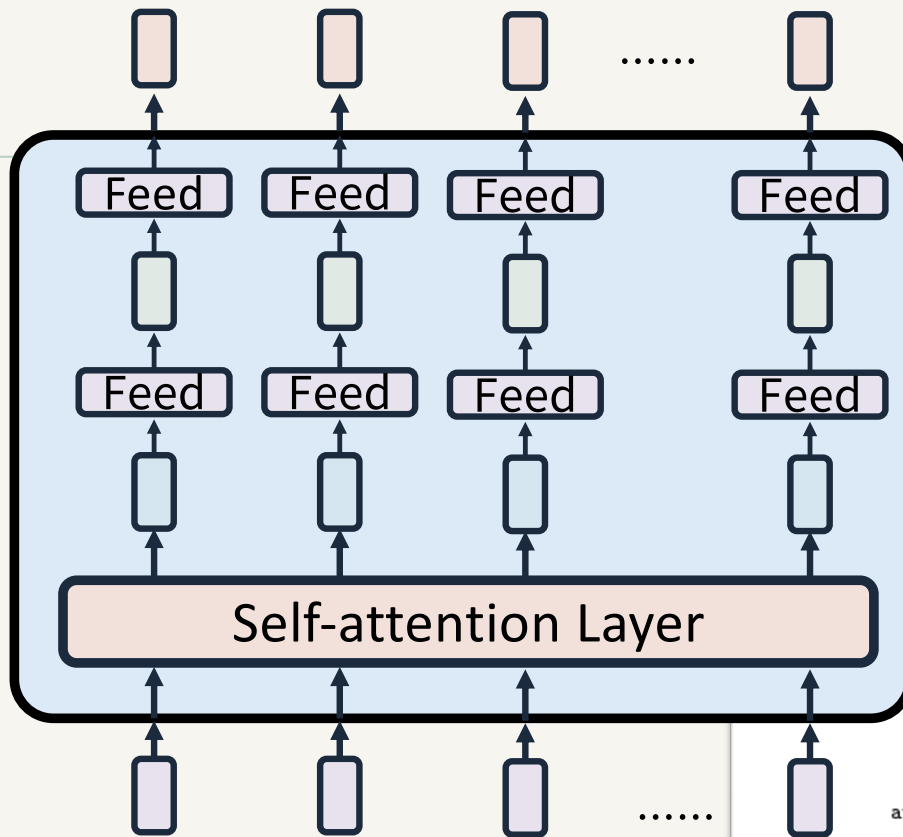
Unboxing LLM



Llama-3.3-70B-Instruct



Layer



Transformer

Attention Is All You Need

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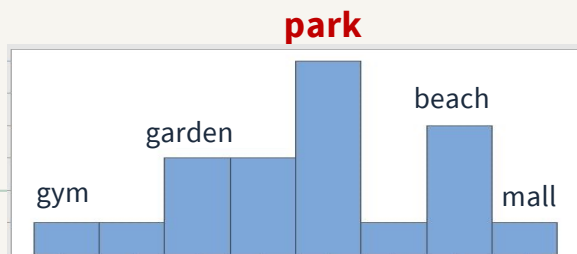
Illia Polosukhin* ‡
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31st Conference on Neural Information Processing Systems (NIPS 2017), Long Beach, CA, USA.

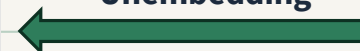
Unboxing LLM



Llama-3.3-70B-Instruct



Unembedding



Layer L

Contextual
Embedding



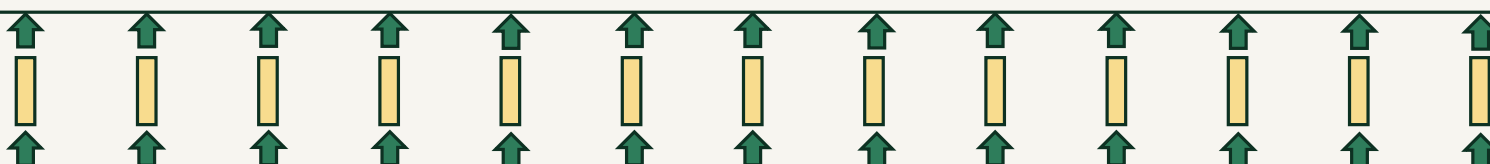
...

Contextual
Embedding



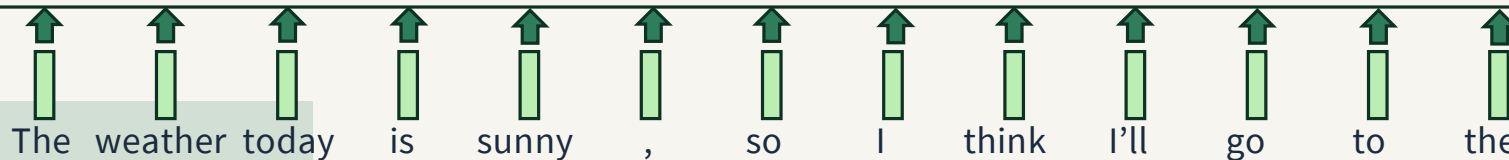
Layer 2

Contextual
Embedding



Layer 1

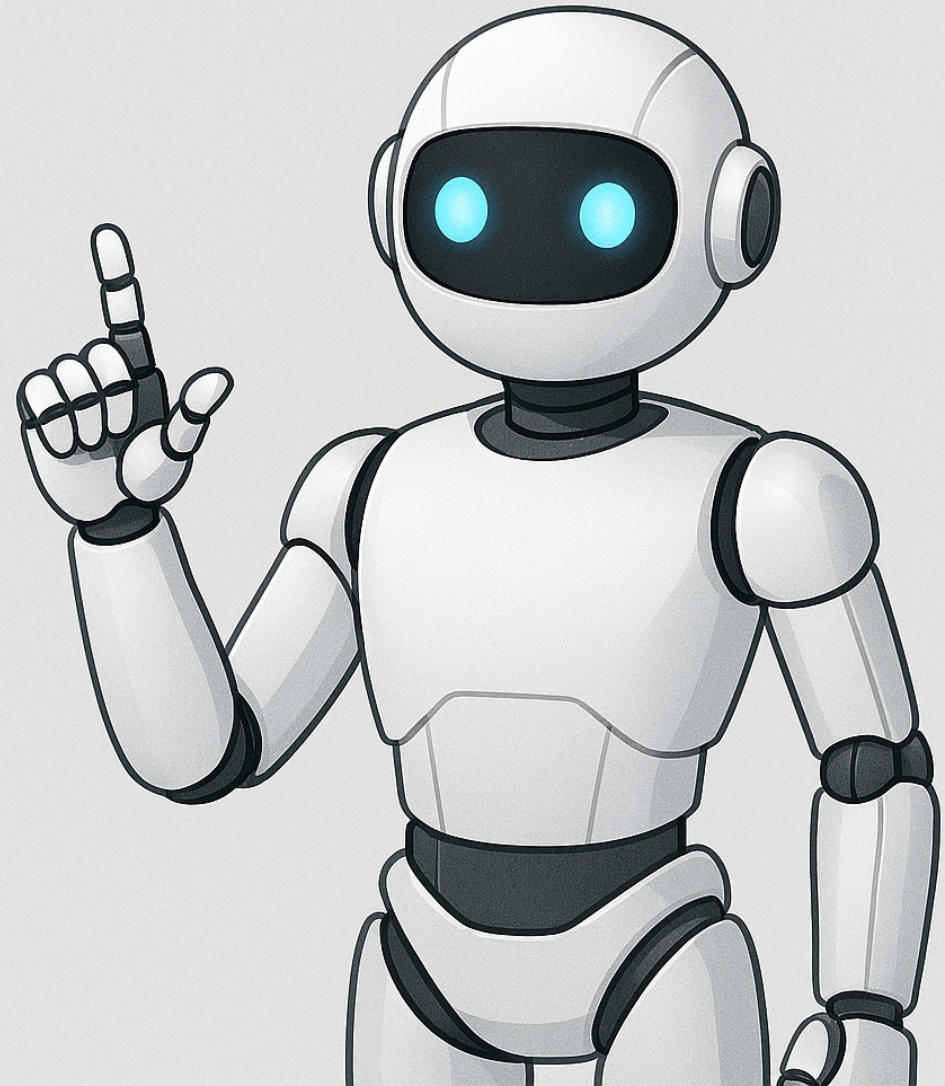
Token
Embedding



Common AI Tasks

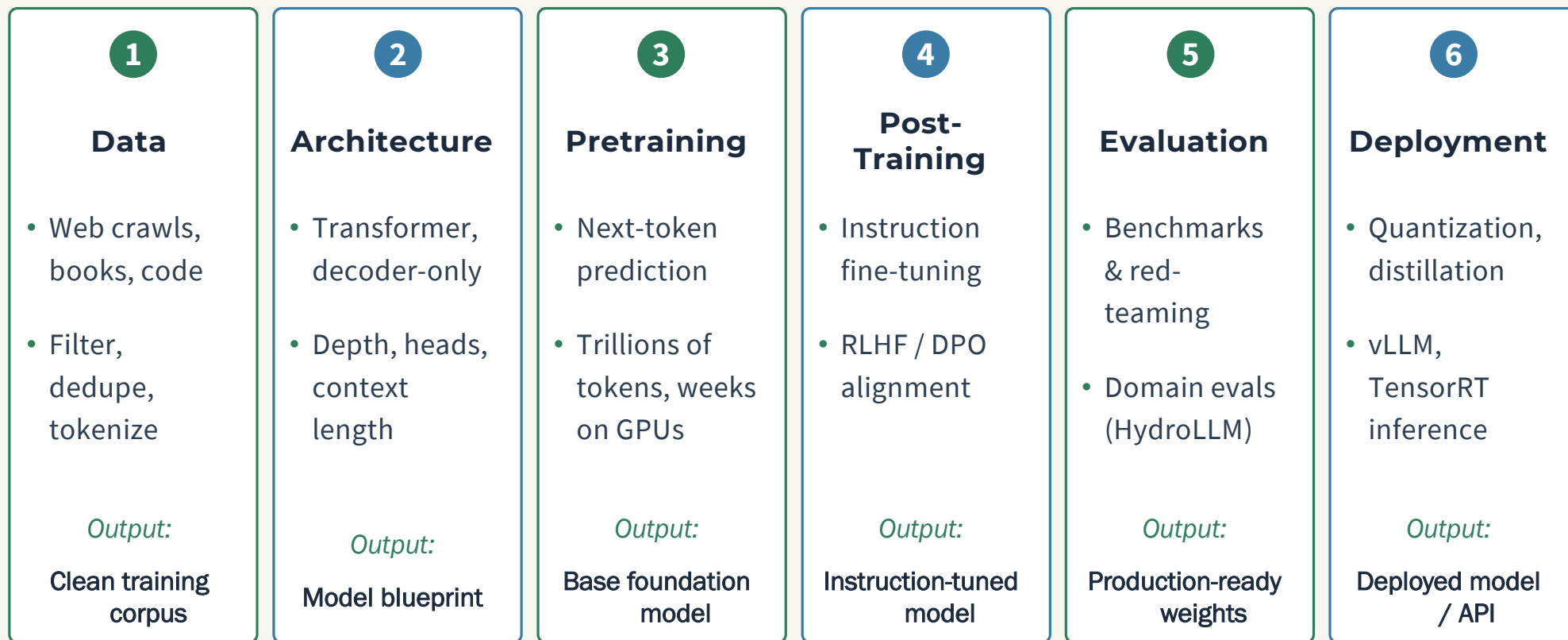
what people are using LLMs for today

- Chatbots and conversation
- Text editing: grammar, proofreading, summarization
- Drafting emails, reports, proposals, slides
- Language translation (incl. live translation)
- Generating audio, images, and video
- Code generation, debugging, technical writing
- Creative writing
- Task and trip planning
- Data analytics and decision support
- Tutoring and learning assistants
- Research: literature review, ideation, synthesis



Recipe for End-to-End LLM Development

six stages from raw data to a deployed model



ranked by capabilities, sized by billion parameters used for training

<https://informationisbeautiful.net/visualizations/the-rise-of-generative-ai-large-language-models-llms-like-chatgpt/>

[illegible]

MMLU = benchmark for measuring LLM capabilities
* = parameters undisclosed // source: LifeArchitect // data

The Model Landscape

Closed / Proprietary

- GPT-5 / GPT-5.5 (OpenAI)
- Claude Opus / Sonnet (Anthropic)
- Gemini (Google)

Best raw performance, but vendor lock-in & privacy concerns

Open-Weight Models

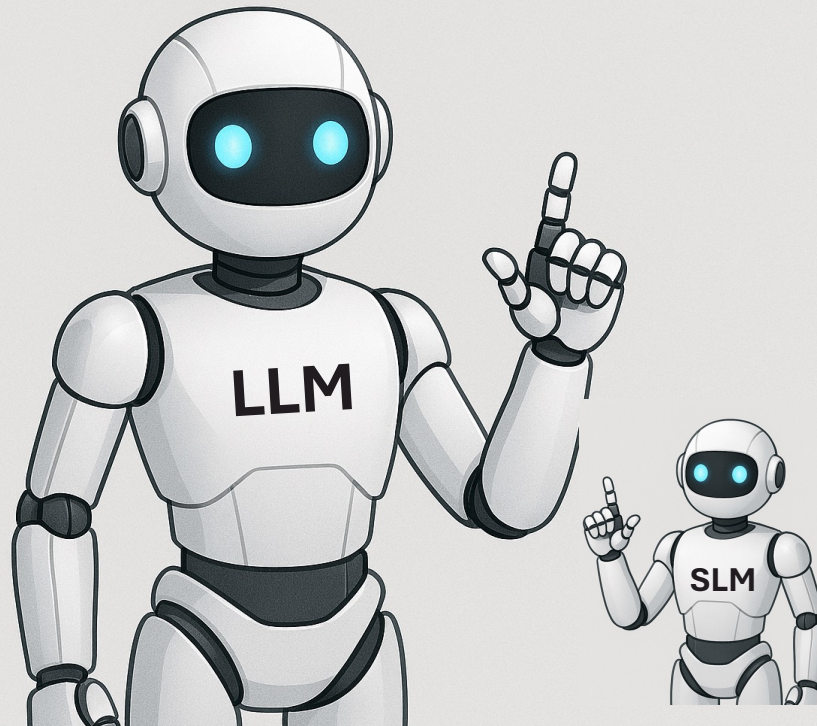
- Llama 4 (Meta) — 10M context
- DeepSeek V4 — top open leader
- Qwen 3.6, Gemma 4, Mistral

Self-host, fine-tune, full data control — gap with proprietary ~3 months

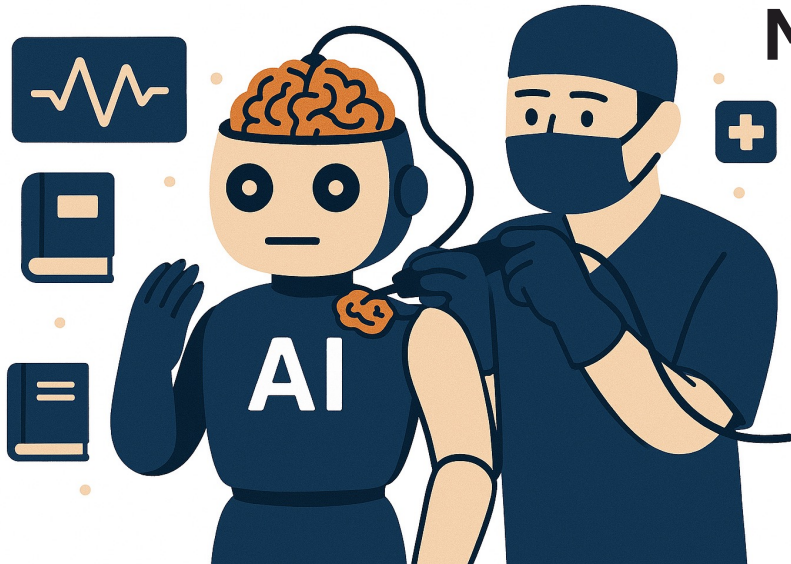
Why this matters: open weights = reproducibility, privacy, and customization for science.

Large vs. Small Language Models

- Around trillions of parameters
- Frontier models; deep language understanding
- Broad knowledge across multiple fields, complex reasoning, open-ended tasks
- Substantial computing resources and operational costs
- Run on high-end GPUs/TPUs, scalable clouds



- Millions to many billions of parameters
- Domain-specific tasks
- Efficient resource utilization
- General computing resources, enterprise, local & edge devices
- Fast, low-latency performance
- More affordable to train or fine-tune
- Fast deployment



Model Fine-tuning = Specialization

Domain Knowledge

Fine-tuning allows LLMs to acquire specialized knowledge in fields like healthcare, law, or finance, enhancing relevance beyond general-purpose capabilities.

Task Optimization

Models are customized for precise tasks such as summarizing medical reports or detecting diseases, resulting in targeted and accurate outputs.

Data Efficiency

Fine-tuning leverages pretrained models, requiring only small amounts of labeled data to adapt effectively, saving time and reducing data collection costs.

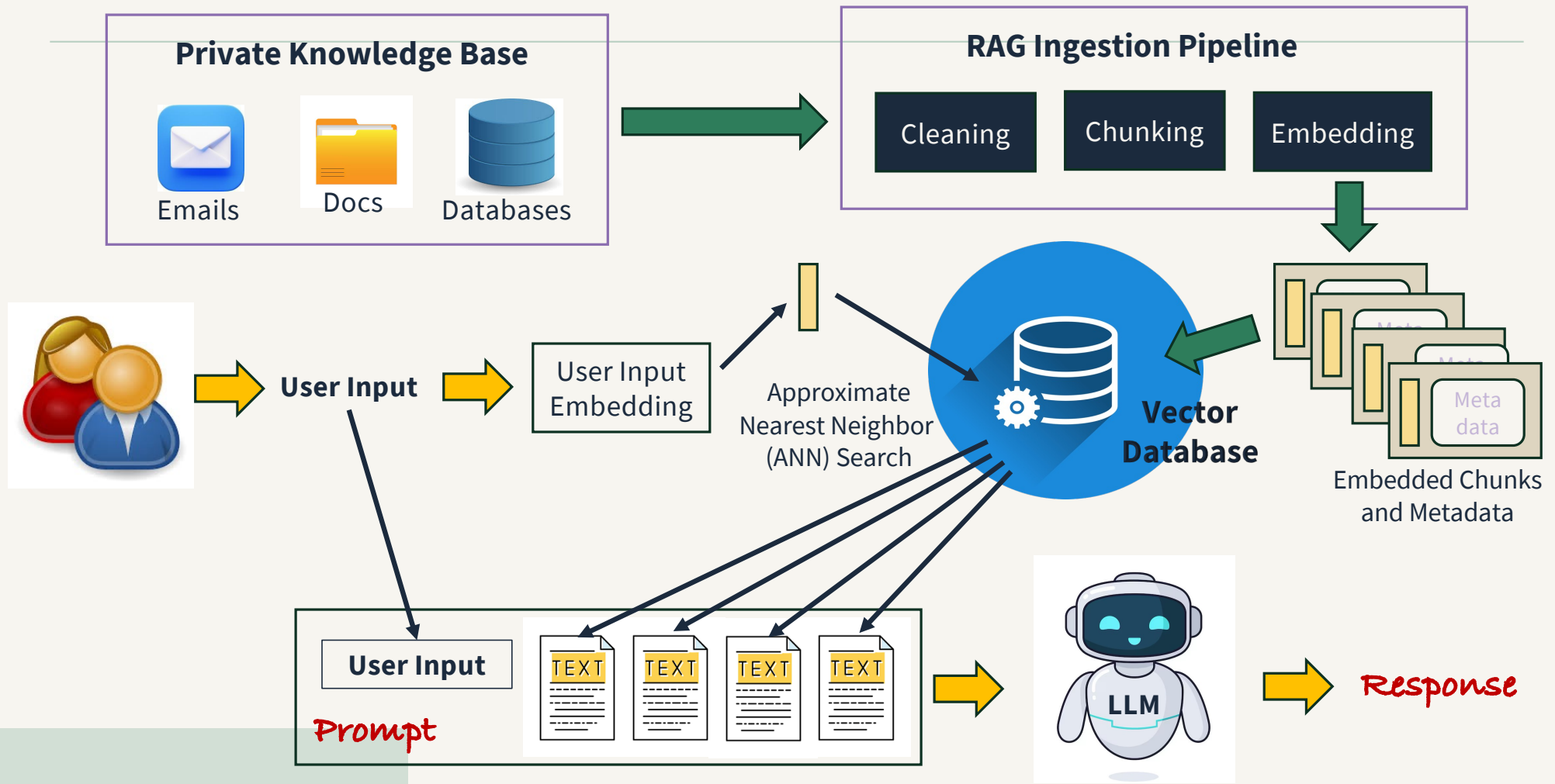
Resource Efficiency

Compared to training from scratch, fine-tuning demands fewer computational resources, enabling faster deployment and more sustainable AI development.

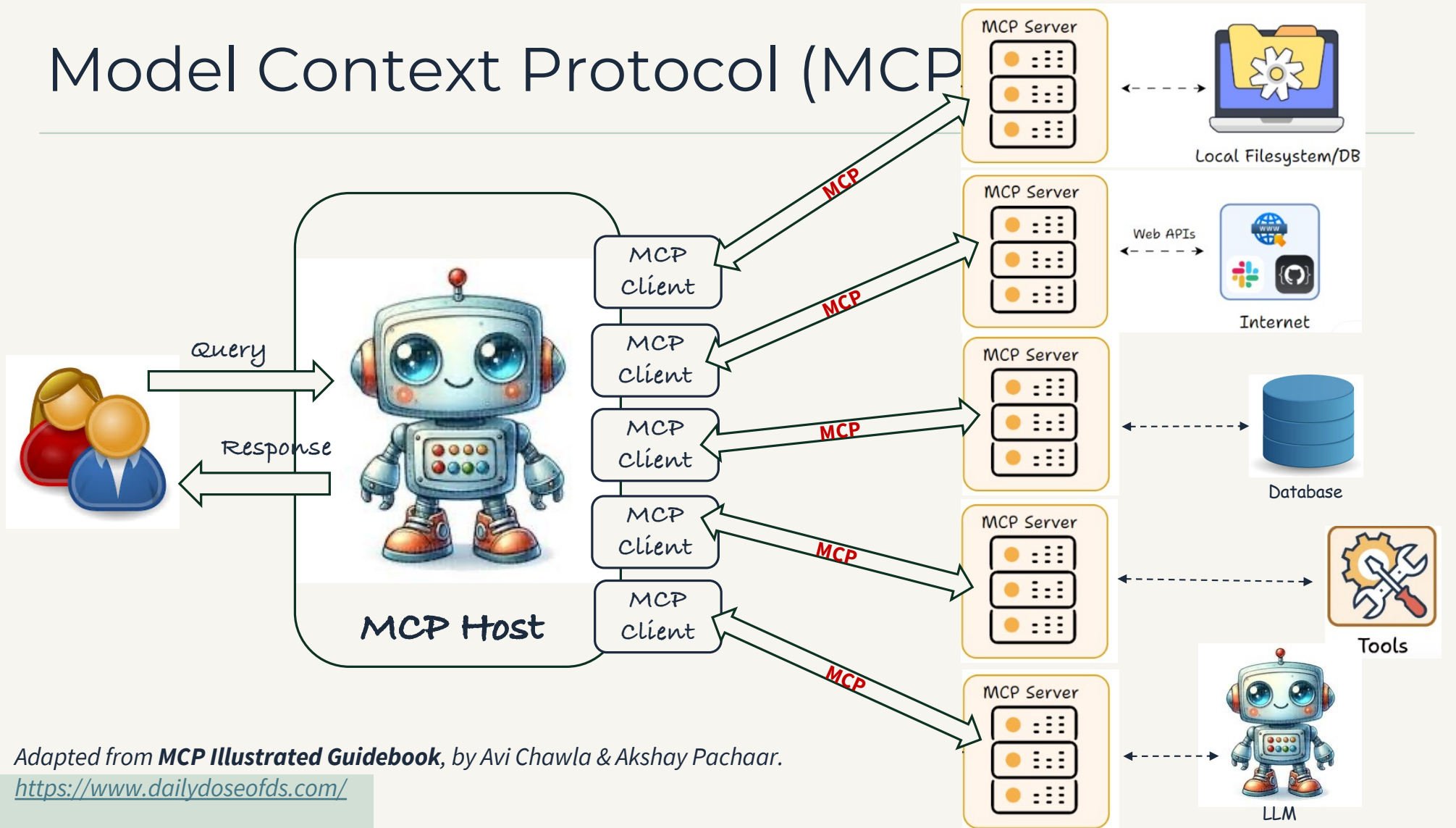
Faster Deployment

Fine-tuned models can be quickly adapted and deployed to meet specific business or research needs, accelerating innovation cycles.

Retrieval Augmented Generation (RAG)



Model Context Protocol (MCP)



Adapted from **MCP Illustrated Guidebook**, by Avi Chawla & Akshay Pachaar.

<https://www.dailydoseofds.com/>

LLM Applications

Hydrology

- **Literature synthesis** — Miao et al. parsed 310k studies, basin-level trends
- **Domain benchmarks & WaterGPTs** — HydroLLM-Benchmark; fine-tuned domain models
- **Conversational interfaces** — RAG over reports; agents drive HEC-HMS, SWAT, MODFLOW

Remote Sensing

- **Geospatial foundation models** — Prithvi (NASA/IBM), SatMAE, Clay; floods, crop type
- **Text-promptable segmentation** — SAM, RemoteCLIP for water bodies, wetlands, blooms
- **Code-gen for GIS pipelines** — Earth Engine, GDAL, rasterio from prompts

Challenges & Responsible Use

What to watch for in scientific work

- **Hallucination** — verify every number and citation; treat outputs as drafts.
- **Reproducibility** — log model, version, temperature, and full prompts.
- **Privacy & disclosure** — keep sensitive data off commercial APIs; declare AI use per journal policy.
- **Energy & equity** — real water/carbon footprint; access is uneven globally.



LLMs are Moving Fast

- **Phase 0:** LLMs as next token predictors (2018-2020)
 - Google BERT, OpenAI's GPT-1/2/3, Google LAMDA, etc.
- **Phase 1:** Finetuned LLM as assistant (2022-2024)
 - ChatGPT, Claude, Gemini
- **Phase 2:** LLM + tool-use as “scoped” agents (2024-2025)
 - Google AI Overviews, LangChain, AutoGen, CrewAI
- **Phase 3:** LLM + tool-use + dynamic tool discovery as “autonomous” agents (2025-2026)



Claude Code



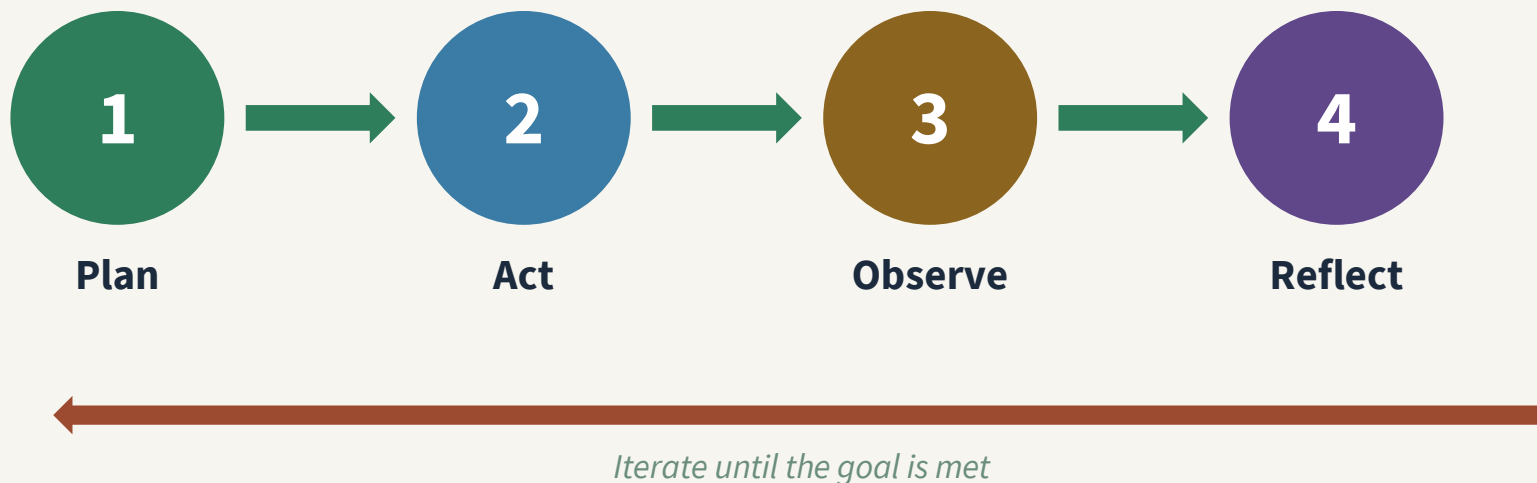
OpenClaw



Hermite Agent

AI Agents: From Chatbots to Doers

LLMs that take actions, not just answer questions



In practice: Claude Code, OpenClaw, ChatGPT agents — connected to tools via MCP, browsers, and shell commands.

Prompt, Context & Harness Engineering

three layers of getting useful behavior out of an LLM

Prompt Engineering

what you say to the model

- **Clear instructions** — role, task, output format
- **Few-shot examples** — show the pattern
- **Chain-of-thought** — "think step by step"

Cheapest lever — try first

Context Engineering

what the model can see

- **Retrieval (RAG)** — pull in your data
- **Memory & history** — what to keep, drop, summarize
- **Window budget** — chunk, rank, compress

Decides what facts the model has

Harness Engineering

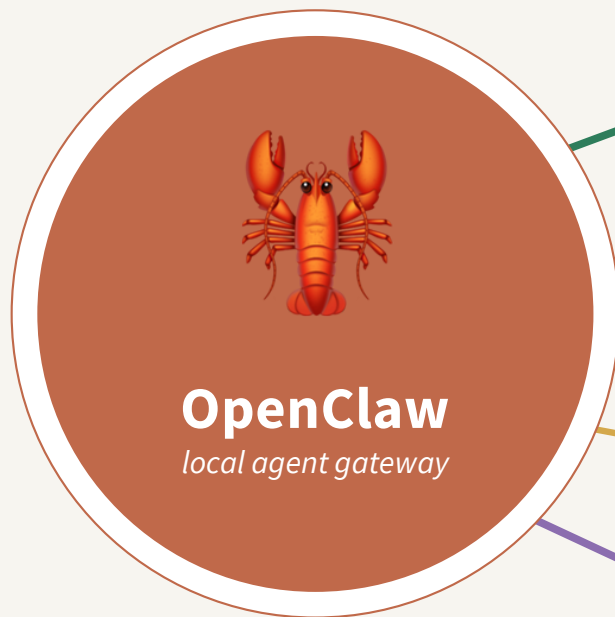
the loop the model runs in

- **Tools & actions** — what the model can do
- **Loops & retries** — verify, plan, recover
- **Guardrails** — permissions, sandboxing, review

Where agents live (Claude Code, MCP)

Spotlight: OpenClaw

a viral local-first AI agent — model-agnostic, open-source



Messaging

WhatsApp, Telegram, Discord, Signal

LLM Backend

Claude, GPT, DeepSeek, or local

Local Tools

Files, shell, browser, scripts

Community Skills

100+ ClawHub plug-in capabilities

Release Nov 2025 • 350,000+ GitHub stars • open-source, model-agnostic, runs locally

Demo

Compound flooding is what happens when multiple flood drivers hit at once — storm surge from the ocean, freshwater from rain-swollen canals, and a water table already sitting near the surface. Each one alone is manageable. Together, they overwhelm a city.

A conversation with Rubi – my OpenClaw research agent.